

Executive Summary

Situation

This project evaluates whether participation in a job-training program causes an increase in post-program earnings. The analysis uses 16,177 observations, including 185 treated individuals and 15,992 comparison individuals. The client needs a causal recommendation rather than a prediction model because the relevant question is not only who earns more, but whether job training itself causes higher earnings.

Complication

A simple comparison may be misleading because training participants and nonparticipants can differ before treatment. The naive OLS estimate is \$-8,498, while the covariate-adjusted OLS estimate is \$699. This difference shows why selection and baseline characteristics matter for the interpretation of the treatment effect.

Resolution

Using Double Machine Learning with demographic and prior-earnings controls, I estimate that job training changes post-program earnings by \$839. The 95 percent confidence interval is [\$-556, \$2,235]. The key identifying assumption is conditional independence: after controlling for observed characteristics, treatment assignment is assumed to be as good as random.

Recommendation

The client should interpret the results cautiously and support job-training programs only with continued evaluation and better baseline data collection. The estimated effect is \$839, but the confidence interval of [\$-556, \$2,235] shows the uncertainty around the estimate. The Streamlit dashboard extends the analysis with what-if scenarios and uncertainty bounds.